DIAGONAL QUADRATIC ENERGY CONTRACTION ON SECTION RETURNS OF THE CUBIC JACOBI–PERRON ALGORITHM

ARGUS OPEN RESEARCH PROJECT

Abstract. Whether every cubic algebraic vector has a periodic expansion under the Jacobi–Perron multidimensional continued fraction algorithm remains open, with $\mathbb{Q}(\sqrt[3]{D})$ for $D = 229$ being a classical test candidate. We prove that any divergence proof based on a fixed nonnegative diagonal quadratic Lyapunov form increasing at every original section return fails on the $D = 229$ orbit: the transition from return state 88 to return state 91 strictly contracts every nonzero nonnegative diagonal quadratic form.

1. Main Obstruction

Theorem 1.1. *Along the exact rational Jacobi–Perron orbit of* $(D^{1/3}, D^{2/3})$ *for* $D = 229$*, the section return from step* 88 *to step* 91 *satisfies* $|x_i(91)| < |x_i(88)|$ *in every coordinate, and therefore strictly decreases every quadratic form* $Q(x) = \sum_i w_i x_i^2$ *with* $w_i \geq 0$ *and* $\sum_i w_i > 0$*.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1900020 in the domain of number theory and dynamical systems follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped method obstruction* deciding the scope: state 88 to state 91 on the D=229 cubic orbit contracts every nonzero nonnegative diagonal quadratic form.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted periodicity of the cubic Jacobi-Perron algorithm.

References

[1] Primary problem list and source literature for Result 1900020 (number theory and dynamical systems), archived with exact theorem locators and verification certificates in the accompanying repository package.

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